Case Study: Health App Implementation at a Hospital

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Overview

Goal: Apply the lessons learned in Module 1 – 3

Case Study: BreatheDeep App

Implementation in a hospital based in New York and in Amsterdam, the Netherlands

BreatheDeep App

- App for patients with asthma and other lung diseases
- Provides recommendations based on Machine Learning: best treatment, medicine use, diet, and exercise, etc.
- Uses data from medical records, patient input, doctor input, remote monitoring data

Steps Towards Trustworthiness

Ensure lawfulness

Identify ethical principles and requirements

Determine implementation methods

Assessment and governance

Ensure Lawfulness

Anna, Hospital Director

Hospital in New York, United States

Existing rules:

- Health Insurance Portability and Accountability Act (HIPAA)
- FDA Medical Device Regulations

AI specific rules:

- Federal or state legislation?

Ensure Lawfulness

Anna, Hospital Director

Hospital in Amsterdam, Netherlands

Existing rules:

- EU General Data Protection Regulation (GDPR)
- EU Medical Device Directive
- Dutch Medical Treatment Contracts Act

AI specific rules:

- Proposal EU AI ACT?

Multidisciplinary Implementation Team

Developer

Doctor / Health Coach

Patient / End User

Identify Ethical Principles and Requirements

Hospital in Amsterdam, Netherlands

Human agency and oversight:

Patients are never obliged to use the app.

Assessment by the doctor must always be ensured.

Privacy and data governance:

Adequate safety measures: two-factor authentication.

Informed consent for exchange and use of hospital data.

Diversity, non-discrimination and fairness

Avoid unfair bias in data set used for training.

The Netherlands: Ethics Guidelines for AI

Determine Implementation Methods

Technical Methods

Explanation methods:

AI Explainability Statement:

https://www.hirevue.com/why-hirevue/ai-explainability-statement

Testing and validating:

Monitoring model during training and development

Testing process designed and performed by a diverse group of people (background, skills, competence)

Red teaming / bug bounties

Determine Implementation Methods

Non-technical Methods

ISO 27001 & 9001

(Standardization)

CODE OF CONDUCT FOR ELECTRONIC DATA EXCHANGE IN HEALTH CARE (NL)

PLAN COMPANY TRAINING
(Education and awareness)

PLAN MEETING WITH PATIENT RIGHTS ASSOCIATION
(Stakeholder participation)

Assessment and Governance

Incorporate assessment process into existing governance

- Address development and deployment at meetings of the hospital board meeting.
- Involve compliance and legal department to keep track of current developments.
- Use the Trustworthy AI assessment list: www.altai.insight-centre.org

Thanks for watching!